EMPLOYEE ATTRITION ANALYSIS

(Data Analytic Project-ReDI School)

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Dataset Overview

- HR Analytic Dataset (Source: Kaggle)

Table 1: Employee details

	EmployeeID	FirstName	LastName	Gender	Age	BusinessTravel	Department	DistanceFromHome (KM)	State	Ethnicity	...	MaritalStatus	Salary	StockOptionLevel	OverTime	HireDate	Attrition	YearsAtCompany	YearsInMostRecentRole
0	3012-1A41	Leonelle	Simco	Female	30	Some Travel	Sales	27	IL	White	...	Divorced	102059	1	No	2012-01-03	No	10	4
1	CBCB-9C9D	Leonerd	Aland	Male	38	Some Travel	Sales	23	CA	White	...	Single	157718	0	Yes	2012-01-04	No	10	6

Table 2: Performance ratings

	PerformanceID	EmployeeID	ReviewDate	EnvironmentSatisfaction	JobSatisfaction	RelationshipSatisfaction	TrainingOpportunitiesWithinYear	TrainingOpportunitiesTaken	WorkLifeBalance	SelfRating	ManagerRating
0	PR01	79F7-78EC	1/2/2013	5	4	5	1	0	4	4	4
1	PR02	B61E-0F26	1/3/2013	5	4	4	1	3	4	4	3
2	PR03	F5E3-48BB	1/3/2013	3	4	5	3	2	3	5	4

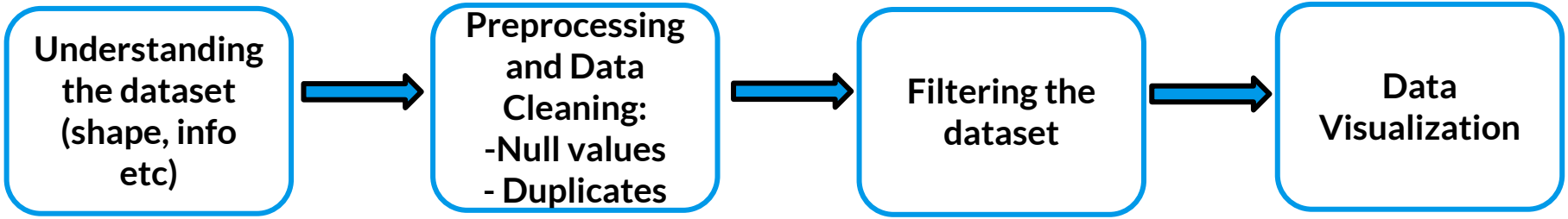
- **Target variable: Attrition (Yes/No)**
- **Features explored:**
  Demographics: *Age, Gender, Education, Marital Status*
  Job-related: *JobRole, Department, JobSatisfaction*
  Organizational: *EnvironmentSatisfaction, WorkLifeBalance*
  Compensation: *Annual Income, PercentSalaryHike*

****Attrition**
Yes - Left the company
No - Stayed

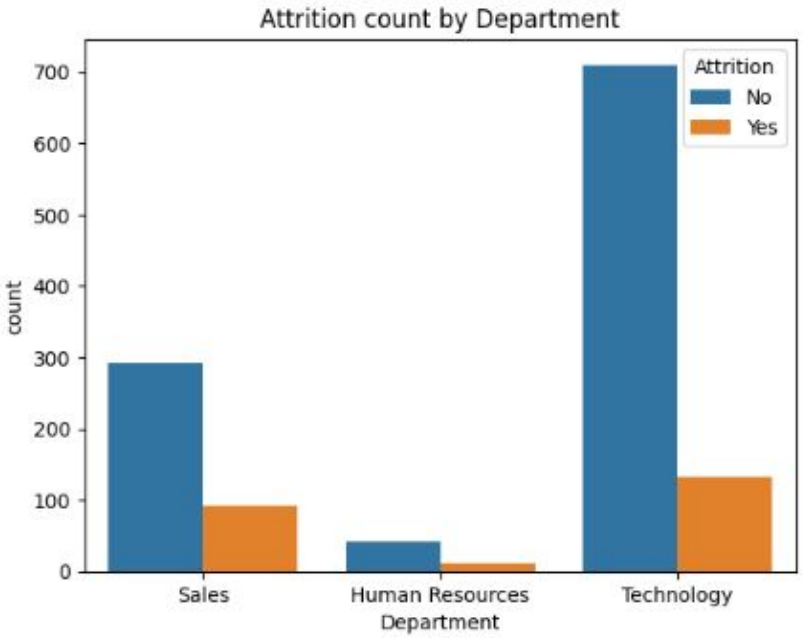
Key Objectives

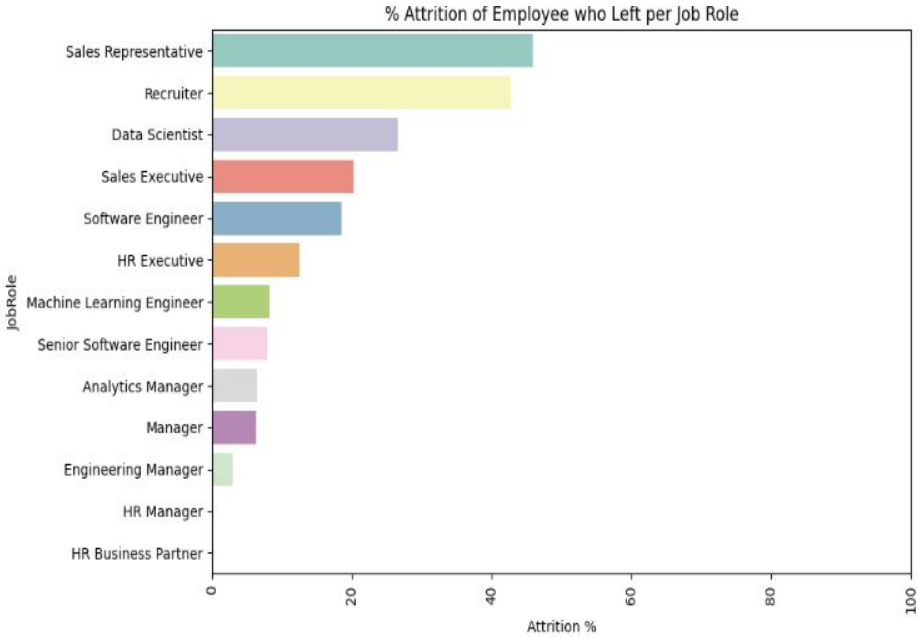
- What contributes to employee attrition (Leaving)?
- Explore relationships between key variables
- Provide insights for company HR policy and retention strategies

Steps: (using python Libraries)



Data Visualization

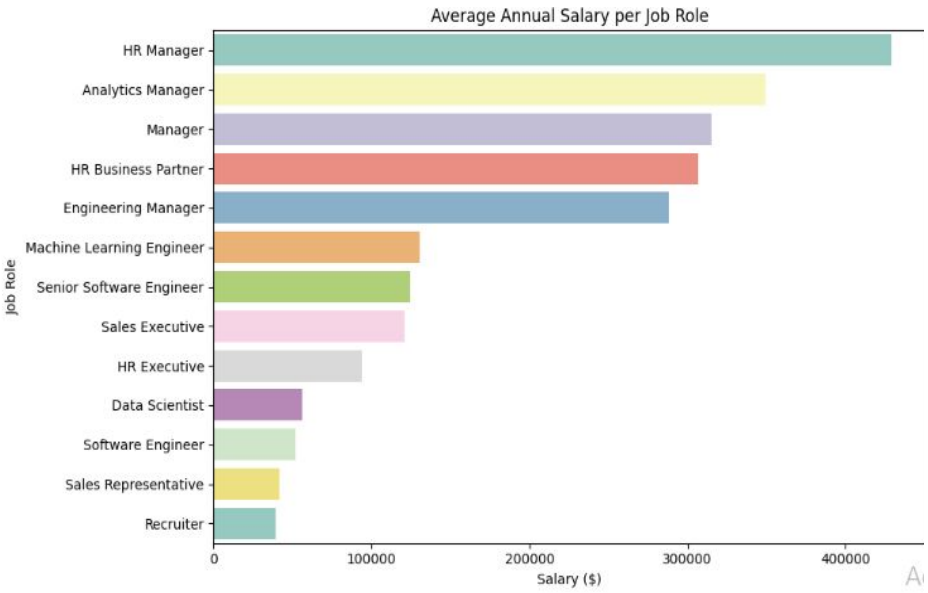


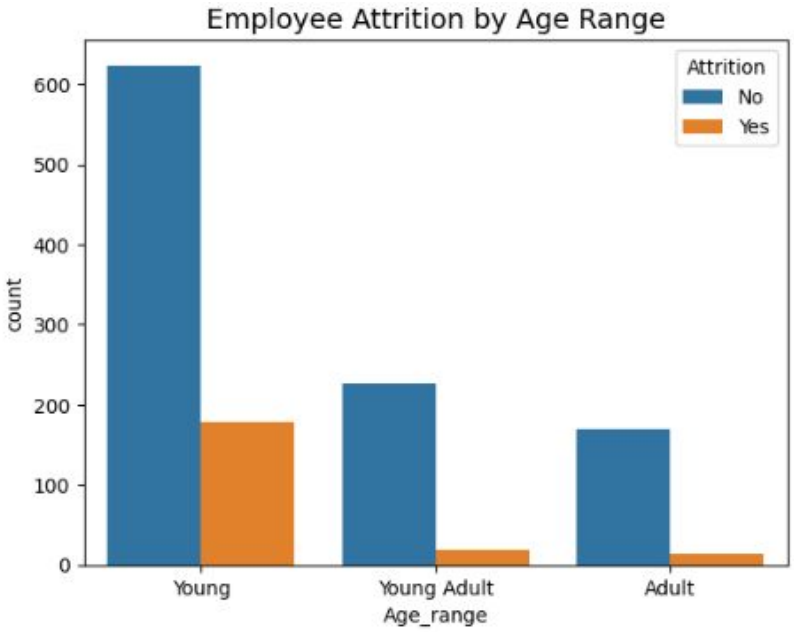


Sales department has a larger share of Sales employees leaving the company

Sales Representative > Recruiter > Data Scientist>Sales Executive

Data Visualization

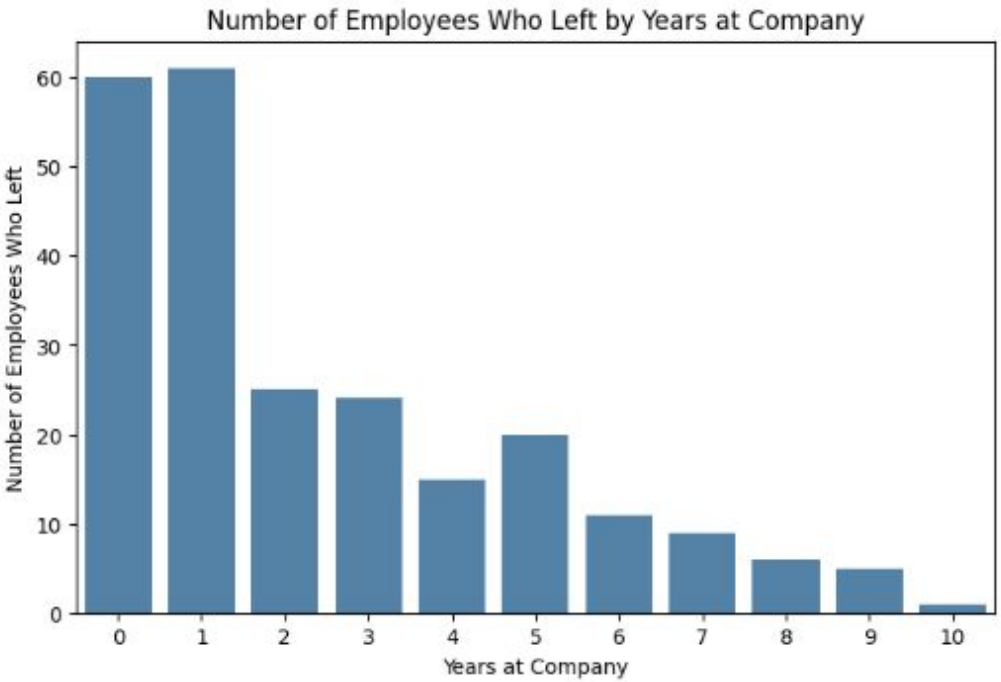




Sales Representative and Recruiter have the lowest annual salary

Young Employees have the highest share of leaving the company

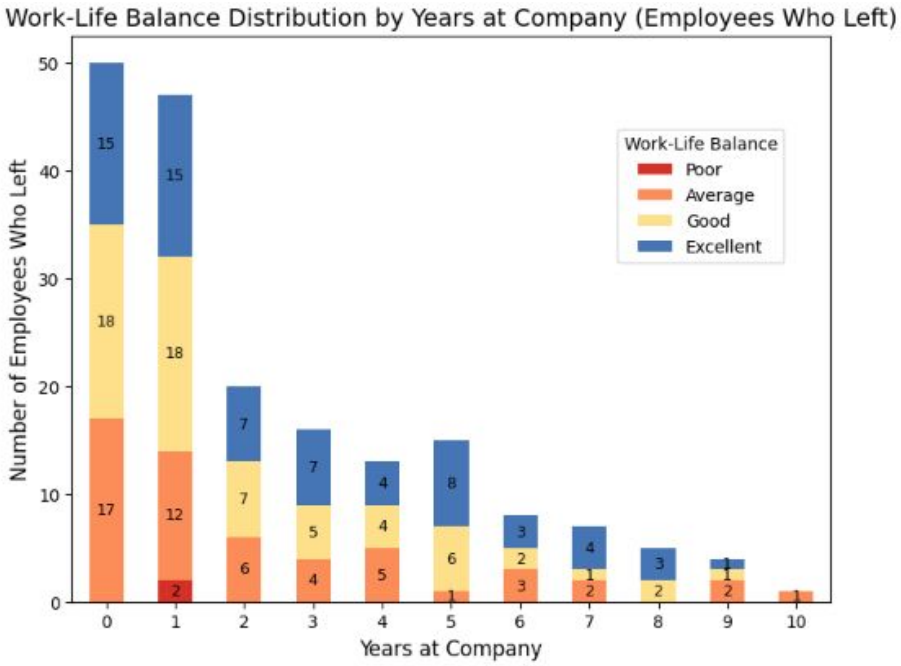
Data Visualization

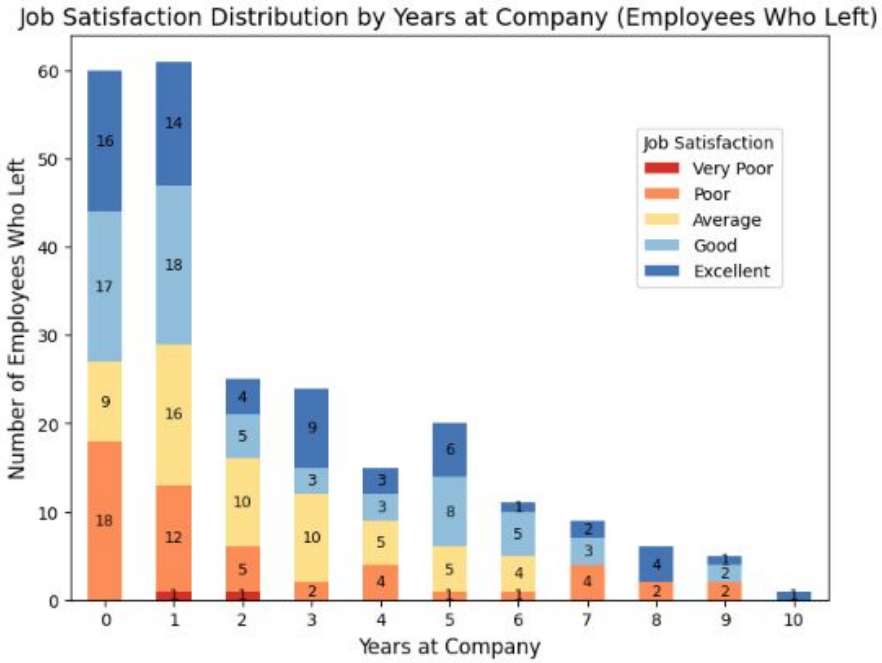


Most Employees Leave more within their first 0–2 Years while very Few Employees Leave After 5+ Years

Most employees leaving in early years are low salary earners

Data Visualization





Work Life Balance and Job Satisfaction are not necessarily the determining factor of employees leaving since high-rating employees still leave.

Key Insights & Summary

- Most employees that have left are among young and low-income employees due:
  - -Desire for faster career growth
  - - Better salary offers elsewhere

- More employees leave within 0-2 years maybe due to:
  - -New hires feel unsupported.
  - -They may not see career growth.
  - -Expectations may not match reality.
  - -Onboarding or early work environment might need improvement.

Recommendation

- Improve work-life balance policies
- Provide competitive salary adjustments
- Conduct regular satisfaction surveys

THANK YOU FOR LISTENING